Example 1.7 If $n(A) = 10$ and $n(A \cap B) = 3$, find $n((A \cap B)' \cap A)$.

Solution:
$(A \cap B)' \cap A = (A' \cup B') \cap A = (A' \cap A) \cup (B' \cap A) = \emptyset \cup (B' \cap A) = (B' \cap A) = A - B$.
So $n((A \cap B)' \cap A) = n(A - B) = n(A) - n(A \cap B) = 7$.

Example 1.8 If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, find $n((A \cup B) \times (A \cap B) \times (A \Delta B))$.

Solution:
We have $n(A \cup B) = 6, n(A \cap B) = 2$ and $n(A \Delta B) = 4$.
So, $n((A \cup B) \times (A \cap B) \times (A \Delta B)) = n(A \cup B) \times n(A \cap B) \times n(A \Delta B) = 6 \times 2 \times 4 = 48$.

Example 1.9 If $\mathscr{P}(A)$ denotes the power set of A, then find $n(\mathscr{P}(\mathscr{P}(\mathscr{P}(\emptyset))))$.

Solution:
Since $\mathscr{P}(\emptyset)$ contains 1 element, $\mathscr{P}(\mathscr{P}(\emptyset))$ contains 2^1 elements and hence $\mathscr{P}(\mathscr{P}(\mathscr{P}(\emptyset)))$ contains 2^2 elements. That is, 4 elements.



Exercise - 1.1

1. Write the following in roster form.
   (i) $\{x \in \mathbb{N} : x^2 < 121$ and x is a prime$\}$.
   (ii) the set of all positive roots of the equation $(x - 1)(x + 1)(x^2 - 1) = 0$.
   (iii) $\{x \in \mathbb{N} : 4x + 9 < 52\}$.
   (iv) $\{x : \frac{x-4}{x+2} = 3, x \in \mathbb{R} - \{-2\}\}$.
2. Write the set $\{-1, 1\}$ in set builder form.
3. State whether the following sets are finite or infinite.
   (i) $\{x \in \mathbb{N} : x$ is an even prime number$\}$.
   (ii) $\{x \in \mathbb{N} : x$ is an odd prime number$\}$.
   (iii) $\{x \in \mathbb{Z} : x$ is even and less than 10$\}$.
   (iv) $\{x \in \mathbb{R} : x$ is a rational number$\}$.
   (v) $\{x \in \mathbb{N} : x$ is a rational number$\}$.
4. By taking suitable sets A, B, C, verify the following results:
   (i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
   (ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
   (iii) $(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$.
   (iv) $C - (B - A) = (C \cap A) \cup (C \cap B')$.
   (v) $(B - A) \cap C = (B \cap C) - A = B \cap (C - A)$.
   (vi) $(B - A) \cup C = (B \cup C) - (A - C)$.
5. Justify the trueness of the statement:
   "An element of a set can never be a subset of itself."
6. If $n(\mathscr{P}(A)) = 1024, n(A \cup B) = 15$ and $n(\mathscr{P}(B)) = 32$, then find $n(A \cap B)$.
7. If $n(A \cap B) = 3$ and $n(A \cup B) = 10$, then find $n(\mathscr{P}(A \Delta B))$.
8. For a set $A, A \times A$ contains 16 elements and two of its elements are $(1, 3)$ and $(0, 2)$. Find the elements of A.